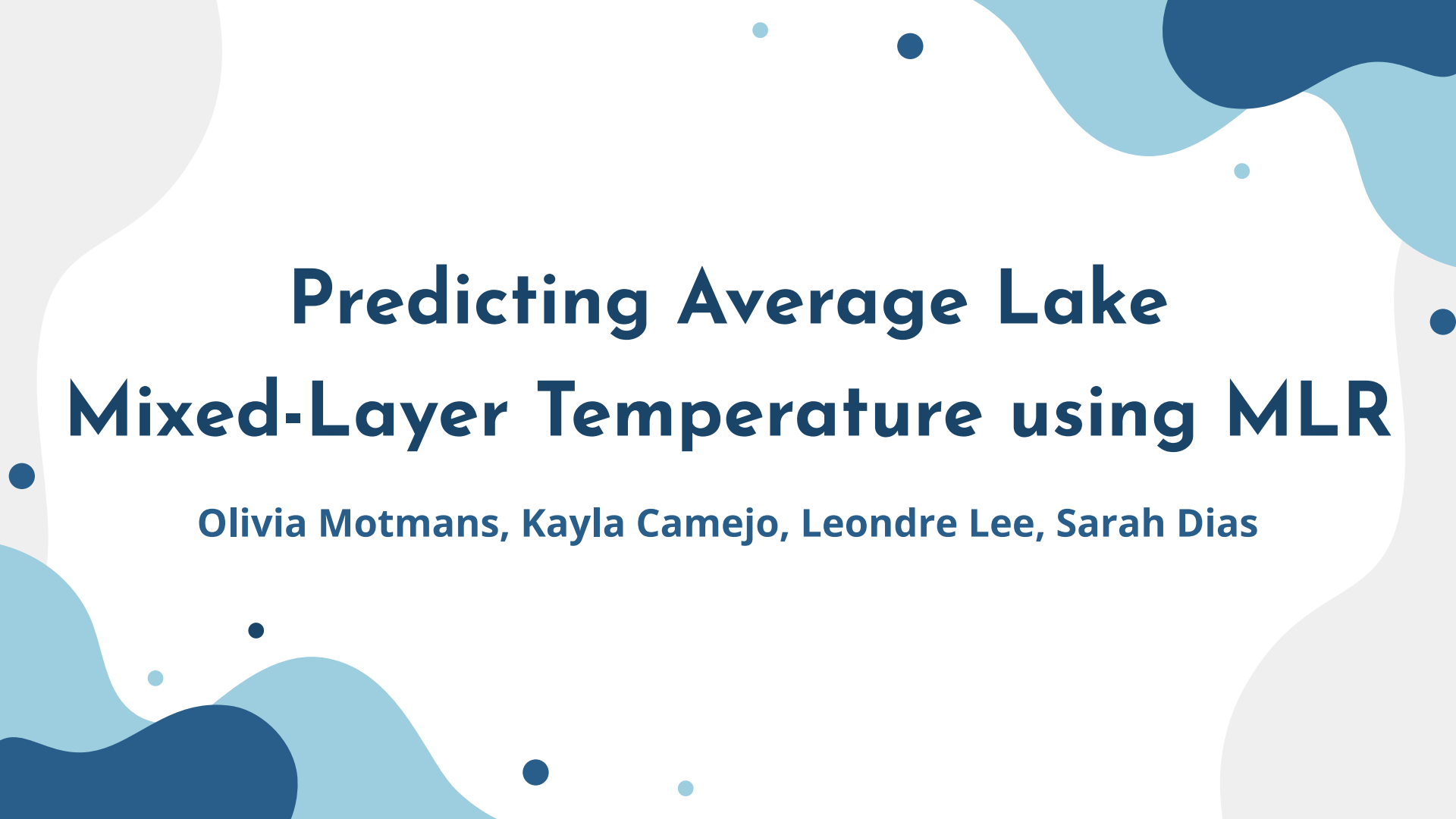




Predicting Average Lake Mixed-Layer Temperature using MLR

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Can we predict lake mixed-layer temperature using atmospheric and lake conditions from Copernicus climate data?

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01

Research question

Predicting Lake Mixed-Layer Temperature

Research Question:

Can we predict average lake mixed-layer temperature using atmospheric and lake conditions from Copernicus climate data?

4 Predictors of Interest

2m temperature

Cloud cover

Lake bottom temperature

Surface solar radiation



Response Variable

Lake mixed-layer temperature

Main idea:

We use multiple linear regression to model lake mixed-layer temperature as a function of 2m air temperature, total cloud cover, lake bottom temperature, and surface solar radiation.

Why this matters:

Lake mixed-layer temperature is important because it reflects how heat is stored and mixed near the lake surface. It can affect ecology, seasonal temperature patterns, and how lakes respond to atmospheric conditions.



02

Our data



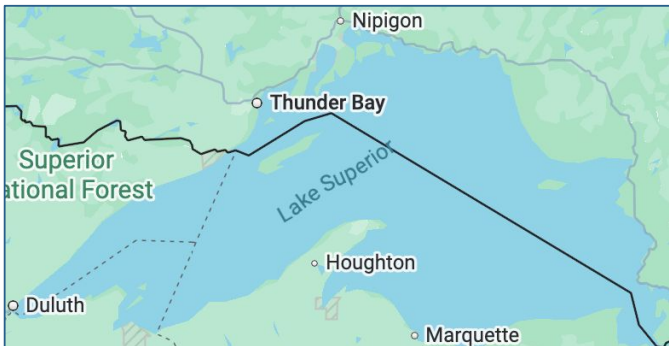
Dataset

ERA5-Land GRIB over Lake Superior region.



Spatial Coverage

Longitude: -92.5 to -84.0
Latitude: 46.0 to 49.2



Data Structure

865 total raster layers
5 repeated variables
173 time layers per variable

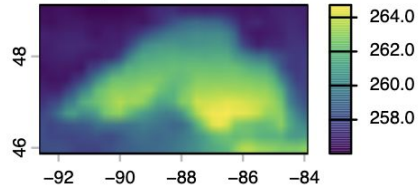


Units

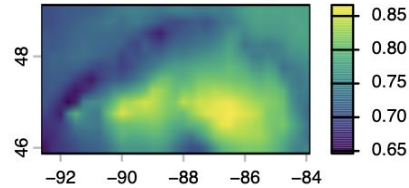
Lake temperatures: Kelvin
2m temperature: Kelvin
Surface solar radiation: $W \cdot s/m^2$
Cloud cover: 0 to 1

First-layer Raster Plots

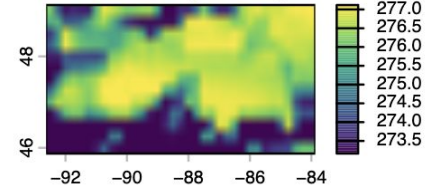
2m Temperature – Layer 1



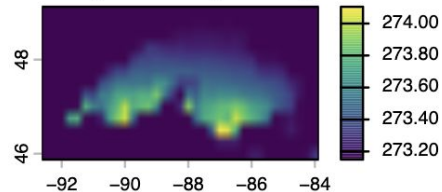
Cloud Cover – Layer 1



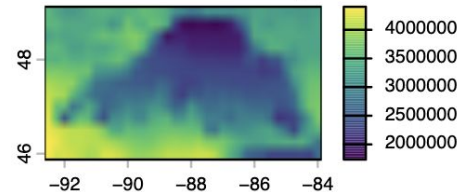
Bottom Temp – Layer 1

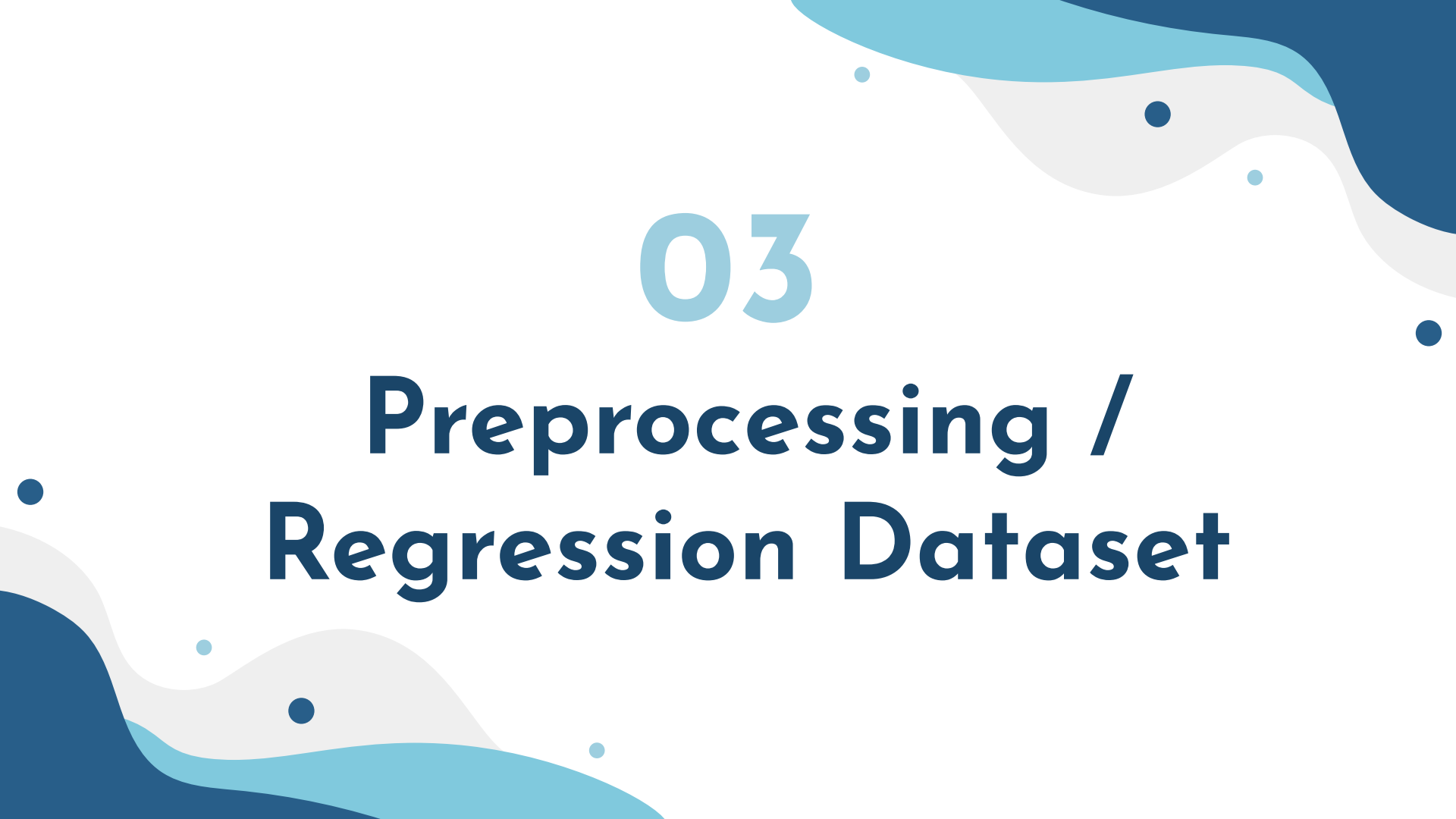


Mix-Layer Temp – Layer 1



Solar Radiation – Layer 1





03

Preprocessing / Regression Dataset

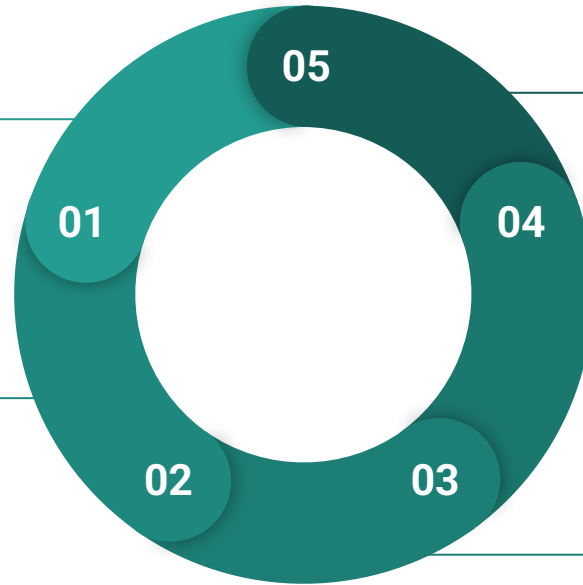
Data Preprocessing: from GRIB file to regression dataset

1. Loaded ERA5-Land GRIB file

Downloaded raw raster data from the Copernicus Climate Data Store

2. Checked data structure

865 total raster layers
5 variables repeated through time
173 layers per variable



5. Built regression table

One row = one time step
Columns = lake mixed-layer temperature + predictor variables

4. Spatially averaged each layer

For each time step, averaged all grid cells over the selected lake region

3. Separated layers by variable

1, 6, 11, ... = 2m temp
2, 7, 12, ... = total cloud cover
3, 8, 13, ... = lake bottom temperature
4, 9, 14, ... = lake mixed-layer temp
5, 10, 15, ... = surface solar radiation

Cleaned Dataset Summary

First 6 rows of dataframe:

	t2m <dbl>	cloud <dbl>	lake_bottom_temp <dbl>	lake_mix_temp <dbl>	solar <dbl>
1	259.7086	0.7575735	275.4674	273.2707	3091967
2	266.6154	0.7906977	275.6958	274.1250	2721120
3	261.4922	0.7747015	275.7066	273.2351	2985005
4	267.4698	0.8846530	275.7041	274.7745	2632552
5	260.7909	0.7577153	275.4593	273.2567	3044463
6	263.9749	0.8525951	275.4139	274.1512	2747321

6 rows

Summary statistics:

t2m	cloud	lake_bottom_temp	lake_mix_temp	solar
Min. :253.3	Min. :0.6292	Min. :275.3	Min. :273.2	Min. :2248638
1st Qu.:260.4	1st Qu.:0.7658	1st Qu.:275.5	1st Qu.:273.3	1st Qu.:2667972
Median :263.8	Median :0.8146	Median :275.6	Median :273.8	Median :2867716
Mean :263.6	Mean :0.8062	Mean :275.6	Mean :273.9	Mean :2879780
3rd Qu.:266.6	3rd Qu.:0.8574	3rd Qu.:275.7	3rd Qu.:274.4	3rd Qu.:3044463
Max. :272.5	Max. :0.9030	Max. :276.3	Max. :275.7	Max. :3758342

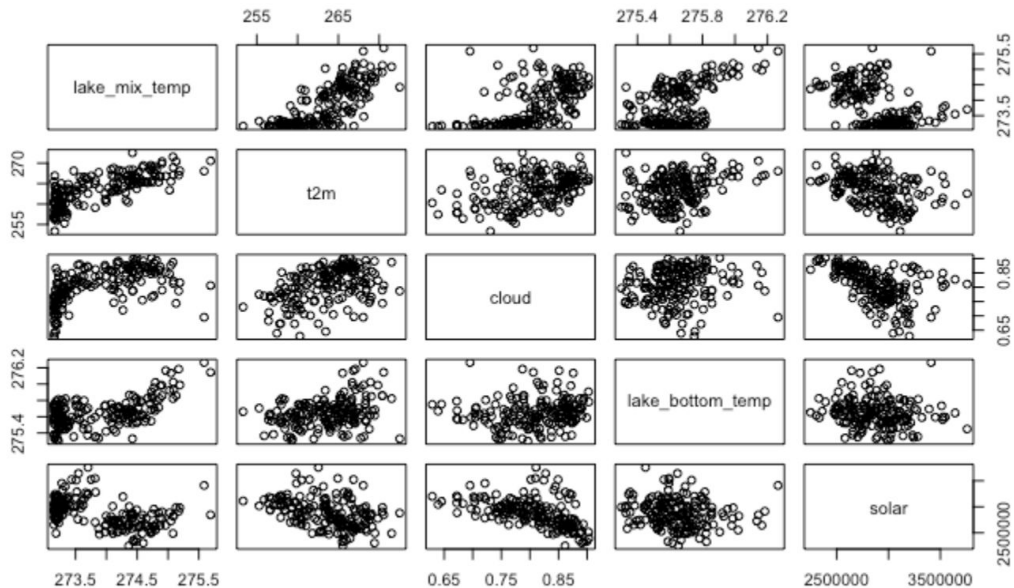


04

Our models

Correlation Between Variables

	lake_mix_temp	t2m	cloud	lake_bottom_temp	solar
lake_mix_temp	1.0000000	0.7534735	0.5417122	0.5666897	-0.4702226
t2m	0.7534735	1.0000000	0.4290397	0.3160510	-0.3851730
cloud	0.5417122	0.4290397	1.0000000	0.1197782	-0.5062691
lake_bottom_temp	0.5666897	0.3160510	0.1197782	1.0000000	-0.1380662
solar	-0.4702226	-0.3851730	-0.5062691	-0.1380662	1.0000000



Baseline Model

We started with a full model with all four predictors:

```
full_model <- lm(lake_mix_temp ~ t2m + cloud + lake_bottom_temp + solar)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.282e+02	4.072e+01	-3.148	0.00195	**
t2m	8.575e-02	7.747e-03	11.069	< 2e-16	***
cloud	2.469e+00	4.995e-01	4.943	1.85e-06	***
lake_bottom_temp	1.372e+00	1.496e-01	9.175	< 2e-16	***
solar	-2.706e-07	1.077e-07	-2.513	0.01291	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3293 on 168 degrees of freedom

Multiple R-squared: 0.7584, Adjusted R-squared: 0.7526

F-statistic: 131.8 on 4 and 168 DF, p-value: < 2.2e-16

As seen in the R output, all variables are seen to be significant.

Two-meter temp (t2m) is the most significant and solar is the least significant predictor.

Baseline Adj. R^2 : **0.7526**

Initial Model Comparison

We compared the full model against simpler models and one interaction model using AIC and adjusted R^2 .

The full model performed better than the simpler models, and adding the cloud:solar interaction improved model fit further.

Model <chr>	Predictors <chr>	AIC <dbl>	Adj_R2 <dbl>
Full model	t2m + cloud + lake_bottom_temp + solar	113.50	0.753
Atmospheric-only model	t2m + cloud + solar	181.77	0.631
Two-temperature model	t2m + lake_bottom_temp	153.91	0.684
Full model + cloud:solar interaction	t2m + cloud + lake_bottom_temp + solar + cloud:solar	107.40	0.763

4 rows

Variable Selection

Although all predictors in our baseline model were seen to be significant, we still wanted to see if there were variables that could be removed.

We tried 3 different variable selection methods:

- Stepwise AIC (both forward and backward)
- Stepwise BIC
- Lasso Regression

Method	Predictors Selected	Score	Notes
Stepwise AIC	All 4	AIC = -379.45	No variable improved AIC on removal
Stepwise BIC	All 4	BIC = -363.68	Heavier penalty, same result
Lasso (lambda.min)	All 4	MSE minimized	Solar var near zero but retained

The results were consistent: All 4 predictors meaningfully add to predictive power and are worth keeping in the model.

Checking for Multicollinearity

Our next step was to check for multicollinearity. We computed **VIF (Variance Inflation Factor)**

t2m	cloud	lake_bottom_temp	solar
1.40	1.47	1.11	1.41

All VIF values were well below the threshold of 5, confirming no multicollinearity concerns.

Even though VIF values do not pass the threshold for concern, interactions are still possible, and are highly plausible for environmental variables. For example,

1. solar $\times$ cloud — Cloud cover directly blocks solar radiation, so the effect of solar on lake temperature likely depends on how cloudy it is
2. t2m $\times$ cloud - Cloud cover affects air temperature (insulating effect), so their combined effect on lake mixed layer temp may not be purely additive
3. t2m $\times$ solar - Higher air temps combined with more solar radiation could have a compounding warming effect on the lake

Testing Interactions

Test for all possible interactions by creating interaction model.

```
interaction_model <- lm(lake_mix_temp ~ (t2m + cloud + lake_bottom_temp + solar)^2)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.300e+04	2.927e+03	4.442	1.64e-05	***
t2m	-5.215e+01	9.971e+00	-5.230	5.18e-07	***
cloud	1.205e+03	9.481e+02	1.271	0.2054	
lake_bottom_temp	-4.640e+01	1.060e+01	-4.378	2.14e-05	***
solar	-6.122e-05	2.047e-04	-0.299	0.7653	
t2m:cloud	-7.707e-03	1.488e-01	-0.052	0.9587	
t2m:lake_bottom_temp	1.899e-01	3.613e-02	5.257	4.56e-07	***
t2m:solar	-3.791e-08	3.499e-08	-1.083	0.2803	
cloud:lake_bottom_temp	-4.314e+00	3.446e+00	-1.252	0.2125	
cloud:solar	-3.928e-06	1.947e-06	-2.018	0.0453	*
lake_bottom_temp:solar	2.691e-07	7.416e-07	0.363	0.7171	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

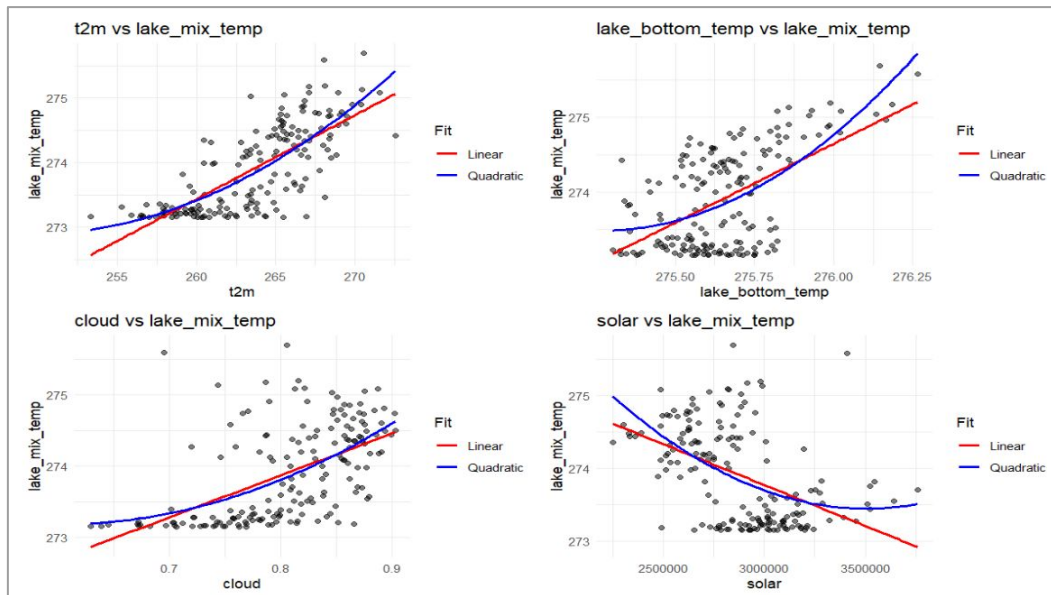
Residual standard error: 0.3012 on 162 degrees of freedom
Multiple R-squared: 0.805, Adjusted R-squared: 0.793
F-statistic: 66.87 on 10 and 162 DF, p-value: < 2.2e-16

Model	Adj R ²	RSE	AIC
Full model (main effects only)	0.7526	0.3293	113.50
All two-way interactions	0.7930	0.3012	88.41
Refined interactions (t2m:lbt + cloud:solar)	0.7926	0.3015	84.94

Interactions **t2m:lake_bottom_temp** (p-value = 4.56e-07) and **cloud:solar** (p-value = 0.0453) were found to be significant. We then added these two interactions to the full model to create a refined model.

Adding Quadratic Terms

We believed that we could make this model better by adding quadratic (squared) terms. We created fitted scatterplots of each variable to see which ones could follow a quadratic pattern. Based on the plots it looks like **t2m** and **lake_bottom_temp** could possibly be better explained with quadratic term.



Call:

```
lm(formula = lake_mix_temp ~ t2m + I(t2m^2) + cloud + I(cloud^2) +  
lake_bottom_temp + I(lake_bottom_temp^2) + solar + I(solar^2) +  
t2m:lake_bottom_temp + cloud:solar, data = model_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.76092	-0.16054	-0.01705	0.15887	0.82306

Coefficients:

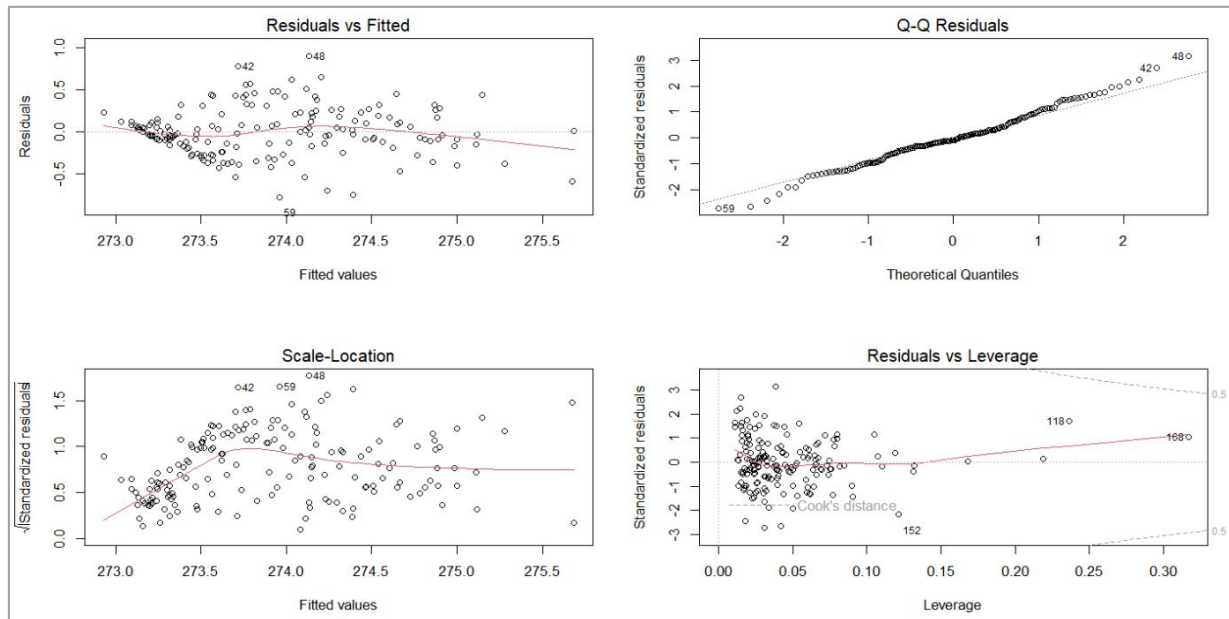
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.246e+04	3.936e+04	2.095	0.03775 *
t2m	-3.308e+01	1.100e+01	-3.009	0.00304 **
I(t2m^2)	4.659e-03	1.419e-03	3.284	0.00126 **
cloud	3.206e+01	1.610e+01	1.992	0.04808 *
I(cloud^2)	-5.341e+00	6.963e+00	-0.767	0.44416
lake_bottom_temp	-5.659e+02	2.902e+02	-1.950	0.05293 .
I(lake_bottom_temp^2)	9.748e-01	5.354e-01	1.821	0.07047 .
solar	7.218e-06	3.063e-06	2.356	0.01966 *
I(solar^2)	-2.927e-13	2.654e-13	-1.103	0.27174
t2m:lake_bottom_temp	1.114e-01	4.006e-02	2.782	0.00605 **
cloud:solar	-6.932e-06	2.390e-06	-2.900	0.00425 **

Quadratic term model shows that the **two meter temperature squared I(t2m^2) variable** is highly significant (p-value = 0.0126)

Updated Model + Model Assumptions

Final Model at this stage becomes: `lm(lake_mix_temp ~ t2m + I(t2m^2) + cloud + lake_bottom_temp + solar + t2m:lake_bottom_temp + cloud:solar, data = model_data)`

Adj. $R^2 = 0.8067$; AIC = 73.6605



The Residuals vs Fitted plot indicates approximate linearity, with only a mild downward trend at higher fitted values suggesting slight residual non-linearity.

Q-Q plot shows heavy tails, indicating nonlinearity. This is a cause for concern.

The Scale-Location plot shows mild heteroscedasticity at lower fitted values, which is acceptable given the sample size of $n = 172$.

The Residuals vs Leverage plot shows no observations beyond Cook's distance, and the Durbin-Watson test confirmed no autocorrelation ($DW = 2.387$, $p = 0.995$), fully satisfying the independence assumption.

Trying Transformations and Investigating Outliers

Transformations we tried:

- Anomaly transformation — recentered data around mean; identical R^2 , no diagnostic improvement
- Box-Cox transformation — flat log-likelihood across all λ values; no transformation superior to original scale

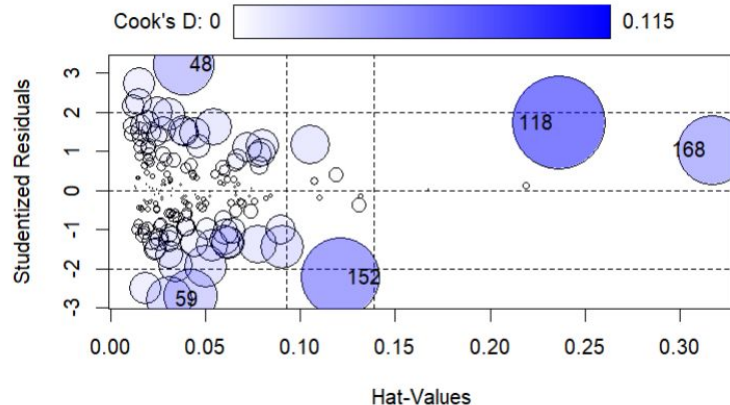
Conclusion: lake_mix_temp is already on the best possible scale. The heavy tails in the Q-Q plot are an inherent property of the climate data. Transformations would just lead to lack of interpretability rather than improved model.

Outliers of Interest and Influence Plot:

- Observation 118 - shown to have large Cook's Distance (shown by the dark blue color) as well as large hat value (above 0.10 threshold)

	t2m <dbl>	cloud <dbl>	lake_bottom_temp <dbl>	lake_mix_temp <dbl>	solar <dbl>
118	268.0533	0.6951427	276.2645	275.5867	3411060

Therefore, we wanted to see what would happen if we removed Observation 118.



Final Model

Final Model: `lm(lake_mix_temp ~ t2m + I(t2m^2) + cloud + lake_bottom_temp + solar + t2m:lake_bottom_temp + cloud:solar, data = model_data[-118,])`

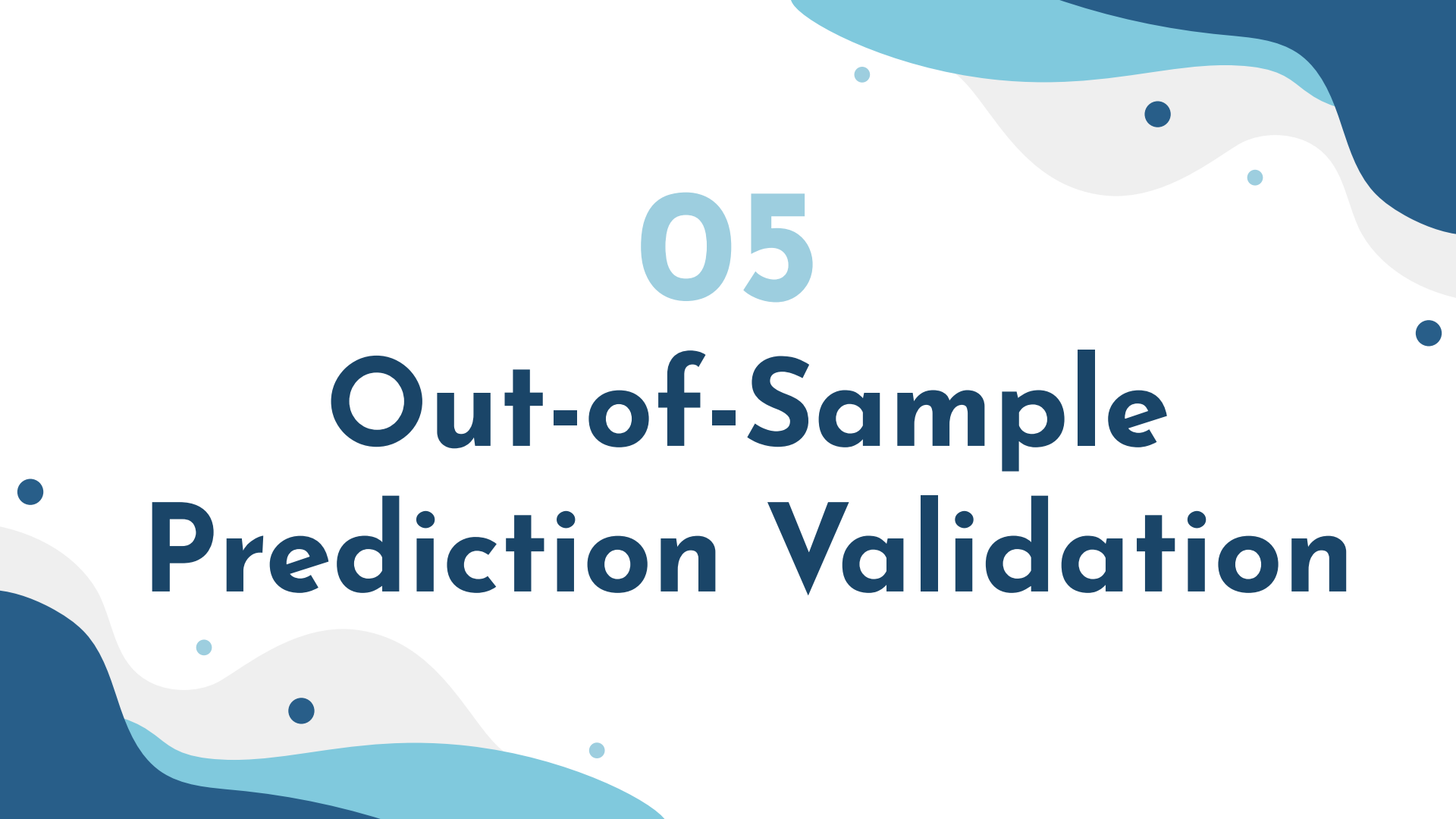
Adjusted AIC: **71.20164 (smallest)!**; Adj. R^2 : **0.8027**; RSE: **0.2892 (smallest)**

Final Coefficients: All coefficients significant at the 0.05 level.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.090e+04	2.544e+03	4.285	3.11e-05	***
t2m	-4.273e+01	9.549e+00	-4.475	1.43e-05	***
I(t2m^2)	5.259e-03	1.399e-03	3.759	0.000237	***
cloud	1.428e+01	5.077e+00	2.812	0.005521	**
lake_bottom_temp	-3.735e+01	9.284e+00	-4.023	8.76e-05	***
solar	2.876e-06	1.442e-06	1.995	0.047748	*
t2m:lake_bottom_temp	1.453e-01	3.501e-02	4.150	5.33e-05	***
cloud:solar	-3.794e-06	1.719e-06	-2.208	0.028662	*

Conclusion: Under standard MLR, the optimal way to predict lake mixed-layer temperature is by using 2 meter temperature (and its quadratic), cloud cover, lake bottom temperature, solar radiation, and the interactions between t2m and lake_bottom_temp and cloud and solar.



05

Out-of-Sample Prediction Validation

Out-of-Sample Validation: 10-Fold Cross-Validation

Nonlinear and Interaction Terms

Allows for the **penalization of overfitting** that can come with our `t2m2`, `cloud:solar` and `t2m:lake_bottom_temp` terms

Autocorrelation between months

Drops entire blocks of time at once to allow for a more honest test without having closely related months to predict from

Outlier impact

Smooths out the impact of possible outliers by averaging performance across larger, data chunks (a concern considering we already found a major outlier (Observation 118))

Grouped K-Fold

Grouped by season to avoid possible information leaks that can come from random splits

10-Fold CV Method and Results

Method

- Resampling pipeline based on **n = 172** observations (Observation 118 left out)
- Partitioned data into **10 folds** while preserving seasonal grouping
- Model is trained on **90%** of historical years and predicting **10%** of block years
- Model executed across all folds using the **OLS method** (lm)

Linear Regression

172 samples
4 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 156, 152, 152, 152, 153, 150, ...

Resampling results:

RMSE	Rsquared	MAE
0.2875919	0.8042114	0.2237911

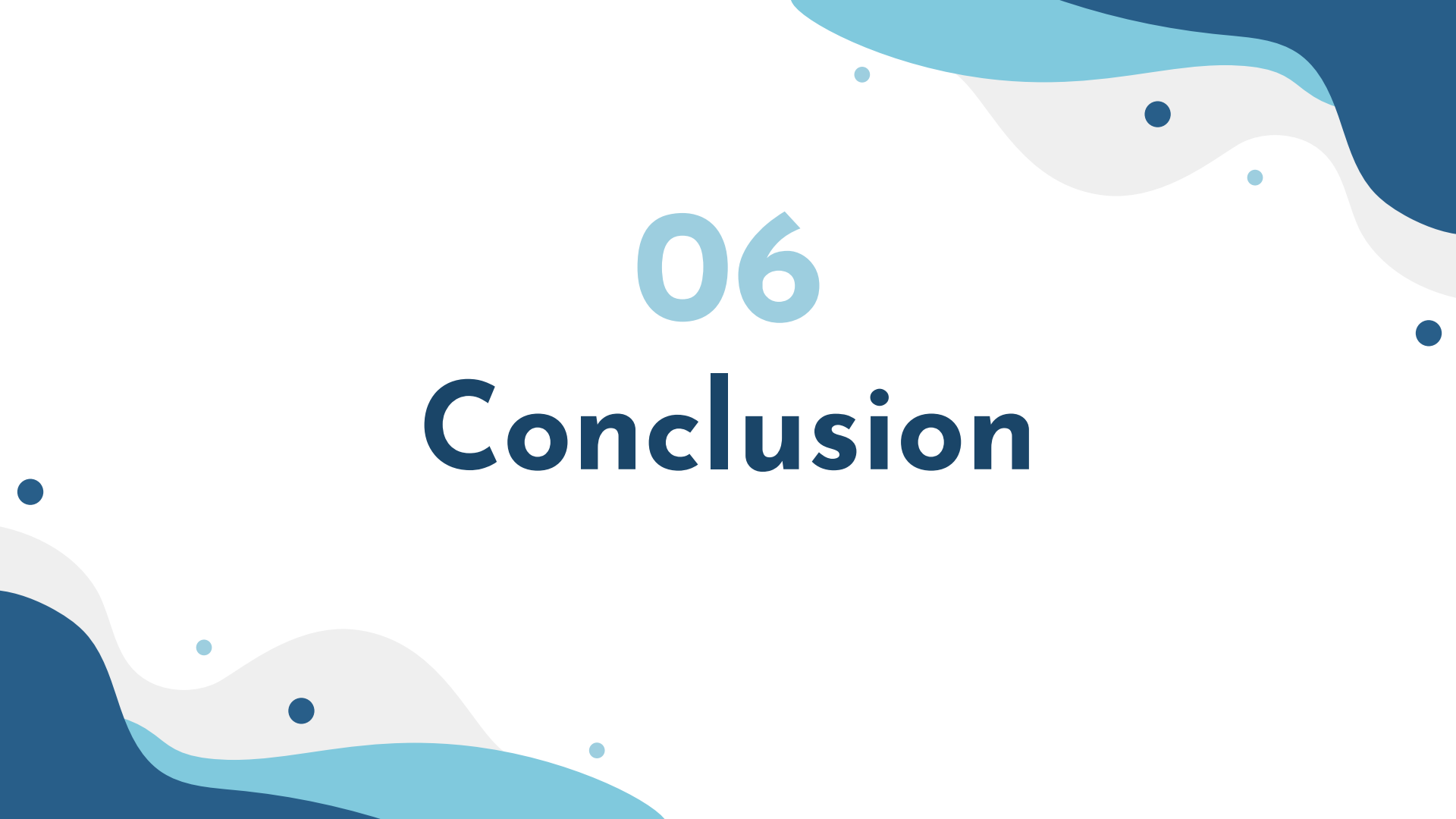
Tuning parameter 'intercept' was held constant at a value of TRUE

Results

- **$R^2 = 0.804$** : Our model is able to explain over 80% of the variance in lake mixed-layer temperature even when looking at entirely new winter seasons!

Error Metrics

- **Root Mean Squared Error (RMSE) = 0.288**
- **Mean Absolute Error (MAE) = 0.224**
Both errors remain low at under a third of a degree Kelvin each



06

Conclusion

Key Takeaways

Captured Feedback

Our framework successfully maps nonlinear air temperature curvature and accounts for cloud-solar and air-lake thermodynamic interactions

Methodological Integrity

We maintained model validity through rigorous variable selection (AIC/BIC/Lasso), multicollinearity verification, Grouped 10-Fold CV, and more



Robust Model

Our final model maintained a roughly 80% r-squared through in-sample and out-of-sample validation



Predictive Precision

The model shows stability and precision across 86 years of winter data with an MAE of 0.22° Celsius



Future Prospects

To further improve our model, we can:

- Expand our data by pulling from global climate drivers such as ENSO, AO, or PDO to capture multi-decadal climate cycles
- Use individual pixel/grid cell level data modeling instead of averaging over a full layer to more accurately represent Lake Superior
- Employ deep learning through machine learning models such as Random Forests to reduce manual labor and detect high-order interactions



Thank you!